

## 0.6 Fields

### Definition 0.6.1 — Field

A set  $F$  is called a **field** if we can define two binary operations addition  $+$ , and multiplication  $\cdot$

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F$$

such that:

- (F1)  $a + b = b + a$
- (F2)  $(a + b) + c = a + (b + c)$
- (F3) There exists  $0 \in F$  such that  $a + 0 = a$ .
- (F4) For every  $a \in F$ , there exists  $-a \in F$  such that  $a + (-a) = 0$ .
- (F5)  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- (F6)  $a \cdot b = b \cdot a$
- (F7) There exists  $1 \in F$  such that  $1 \neq 0$  and  $a \cdot 1 = a$ .
- (F8) For every  $a \in F \setminus \{0\}$ , there exists  $a^{-1} \in F$  such that  $a \cdot a^{-1} = 1$ .
- (F9)  $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

*Remark.* We can shorten “ $F$  is a field with operations  $+$  and  $\cdot$ ” to “ $(F, +, \cdot)$  is a field”. Moreover when the context of  $+$  and  $\cdot$  is clear, we simply say “ $F$  is a field”, which we will do most of the time in this book.

### Example. (Examples of Fields)

The following sets are examples (and non-examples) of fields:

1.  $\mathbb{R}$  is a field.
2.  $\mathbb{C}$  is a field.
3.  $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$  is a field.
4.  $\mathbb{Z}$  is **not** a field since **(F8)** fails.
5.  $M_n(\mathbb{R})$  is **not** a field since **(F6)** fails.

### Example. (Finite Fields)

Let  $\mathbb{F}_n$  be the finite field with  $n$  elements. Then  $(\mathbb{F}_2, +, \cdot)$  is defined to be:

$$\mathbb{F}_2 = \{0, 1\}, \quad \begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array}, \quad \begin{array}{c|cc} \cdot & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}.$$

Then one can verify that  $(\mathbb{F}_2, +, \cdot)$  is a field.